tvst

Retina

Comparison of Changes in Retinal Vascular Density and Thickness After Using Low-Level Red Light and 0.01% Atropine in Premyopic Children

Lili Shang[1,*], Shasha Gao[1,*], Weiqun Wang[1], Minghang Chang[1], Nana Ma[1], Congcong Huang[1], Shiao Yu[1], Ming Wang[2], and Aicun Fu[1]

[1] Department of Ophthalmology, the First Affiliated Hospital of Zhengzhou University, Zhengzhou, China
[2] Department of Optometry, University of Auckland, Auckland, New Zealand

Correspondence: Aicun Fu, Department of Ophthalmology, the First Affiliated Hospital of Zhengzhou University, Zhengzhou, Henan 450052, China.
e-mail: fuaicun2019@qq.com

Received: February 22, 2024
Accepted: May 24, 2024
Published: June 28, 2024

Keywords: low-level red-light; 0.01% atropine; retinal vascular density; retinal thickness

Citation: Shang L, Gao S, Wang W, Chang M, Ma N, Huang C, Yu S, Wang M, Fu A. Comparison of changes in retinal vascular density and thickness after using low-level red light and 0.01% atropine in premyopic children. Transl Vis Sci Technol. 2024;13(6):23, https://doi.org/10.1167/tvst.13.6.23

Purpose: To compare changes in superficial retinal vascular density (SRVD), deep retinal vascular density (DRVD), and retinal thickness (RT) of the macular zone after repeated low-level red light (RLRL) and 0.01% atropine exposure in premyopic schoolchildren.

Methods: Prospective randomized trial. Sixty-nine schoolchildren with cycloplegic refraction >-0.75 D and ≤ 0.50 D were randomly assigned to RLRL and 0.01% atropine groups. SRVD, DRVD, and RT were measured using swept-source optical coherence tomography at baseline and six months. The macular zone was divided into three concentric rings (fovea, parafovea, and perifovea) using the Early Treatment Diabetic Retinopathy Study.

Results: After six months, the whole, parafoveal, and perifoveal SRVD significantly increased in the two groups (all $P < 0.05$). Multivariate regression analyses showed that none of these changes varied significantly between the two groups (all $P > 0.05$), whereas foveal SRVD remained stable in both groups (all $P > 0.05$). In the RLRL group, the whole and perifoveal DRVD increased significantly (all $P < 0.05$), whereas no statistical difference was observed in the foveal and parafoveal DRVD. DRVD remained stable in the 0.01% atropine group (all $P > 0.05$). No significant differences were observed in RT changes between the two groups (all $P > 0.05$). In comparison, there were no significant changes in SRVD, DRVD, or RT after six months in the placebo group in our previous study.

Conclusions: SRVD increased similarly in the RLRL and 0.01% atropine groups, whereas DRVD increased only in the former group. There were no significant RT changes in either group after six months of treatment in premyopic schoolchildren.

Translational Relevance: This research observed the effects of low-level red light and 0.01% atropine on retinal vasculature, offering valuable insights into myopia progression prevention.

Introduction

The incidences of myopia and high myopia have gradually increased. It is estimated that by 2050, a total of 4758 million people will have myopia (49.8% of the world's population), and 938 million people will have high myopia (9.8% of the world's population).[1] High myopia can lead to severe complications. Therefore there is an urgent need to control myopia progression.

Many studies showed that low-dose atropine could effectively delay the progression of myopia in children.[2,3] Additionally, repeated low-level red light (RLRL) is a newly emerging method to effectively control myopia progression.[4,5] However, the

Copyright 2024 The Authors

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License.

mechanisms by which these two methods delay myopia progression remain unclear. Several studies have explored the possible mechanism of action of 0.01% atropine and RLRL on choroidal parameters, such as increased choroidal thickness and blood perfusion after using 0.01% atropine and RLRL.[5–8] Notably, retinal hypoxia may play an important role in the development of myopia.[9] The retina processes optical defocus information and generates a multilevel signaling cascade from the retina to the choroid and sclera.[10] However, only a few studies have examined changes in retinal parameters after using RLRL and 0.01% atropine.

Currently, the results of studies on the effect of RLRL on retinal vascular density (RVD) are inconsistent. Yang et al.[11] showed that RVD increased for five minutes and decreased from five minutes to one hour after using RLRL in myopic schoolchildren. However, Zhao et al.[12] reported that RVD remained unchanged after using RLRL for four weeks. Additionally, only one study from Wang et al.[13] found no change in RVD after using 0.01% atropine for three months in schoolchildren with myopia. Therefore more studies are needed to verify the effect of 0.01% atropine on retinal parameters. This six-month prospective study compared changes in RVD and RT in the macular zone after RLRL and 0.01% atropine in premyopic schoolchildren. This study may provide theoretical support for the mechanism of action of RLRL and 0.01% atropine administration in controlling myopic shift and axial elongation.

Methods

Study Design

This was a prospective, randomized, two-period crossover trial. The inclusion criteria were as follows: aged six to 12 years with cycloplegic spherical equivalent refraction (SER) > -0.75 and ≤ 0.50 diopters (D), astigmatism < 1.00 D, and monocular best corrected visual acuity of 5.0 or better (5-grade recording method) in both eyes. The five-grade recording method was obtained using the formula: L = 5 – LogMar (logarithm of the minimum angle of resolution). The children were enrolled from the First Affiliated Hospital of Zhengzhou University between July 2021 and September 2021. Children with ocular or systemic diseases or previous myopia control treatment were excluded. All children were randomized at an allocation ratio of 1:1 to the RLRL and 0.01% atropine groups. The crossover design specified two six-month treatment periods separated by a one-month wash-out period. The current study reported the seven-month results. Additionally, the data of placebo group was obtained from our previous study that used the same study protocol, population, and inclusion criteria, and similar demographic and clinical parameters compared to current study.[14]

The study protocol was approved by the Human Ethics Committee of the First Affiliated Hospital of Zhengzhou University (grant number 2021-KY-0399-003). This trial adhered to the tenets of the Declaration of Helsinki and was registered in the Chinese Clinical Trial Registry (ChiCTR2100048406). Verbal assent was obtained from all participants, and written informed consent was obtained from their parents or legal guardians.

RLRL and 0.01% Atropine Eye Drops

The RLRL device (Leshi Yangguang; Chenzhou Eye Care Health Technology Co., Ltd., Hunan, China) emitted diffuse red light at 650 ± 10 nm. At the exit peephole, the diameter of the light spot was 10 ± 2 mm, and the light source energy was approximately 0.85 mW. Children in the RLRL group received RLRL twice daily at home for three minutes per session, with a minimum interval of four hours between sessions. Treatment adherence was recorded using the built-in automated diary function in the RLRL instrument. Children and their parents logged in the RLRL device using assigned accounts to initiate and complete the treatment sessions. The diary function means that the RLRL device can automatically record the date and time of treatment and upload them to the online management platform. On this management platform, the experimenter can view the children's name, identity document, refraction, axial length and fundus images, and regularly check the use time of the instrument. If a child completes less than two sessions per day, the management platform will sent a brief mobile reminder message to the parent to promote treatment compliance. RLRL, with a ratio of less than 90%, indicated poor treatment compliance. Detailed descriptions of 0.01% atropine eye drops have been published elsewhere and are briefly described here,[14,15] 0.01% atropine eye drops were prepared by diluting atropine sulfate powder (Shaoxing Minsheng Medical Co., Ltd., Zhejiang, China) with normal saline solution under sterile conditions and adding a preservative (0.3 mg/mL ethylparaben). Children in the 0.01% atropine group received eye drops after the first examination and once every night. Treatment compliance was considered poor if any bottle containing the remaining eye

drops exceeded 10% of the total amount in each bottle.

RVD and RT Measurements

All participants underwent swept-source optical coherence tomography angiography (SS-OCTA) (VG200S; SVision Imaging, Henan, China). This instrument contained a swept-source laser with a central wavelength of 1050 nm and a scan rate of 200,000 A-scans/s. A detailed description of the basic principles of the SS-OCTA instrument has been published elsewhere.[16,17] RVD and RT were automatically assessed using built-in software (software version: v1.32.9). RVD was obtained by dividing the area covered by the blood flow signal by the measurement area and converting the result into a percentage. RVD was segmented into superficial RVD (SRVD) and deep RVD (DRVD). The SRVD and DRVD were delineated from 5 µm above the inner limiting membrane to 25 µm below the lower border of the inner nuclear layer. Segmentation between the SRVD and DRVD was set in the inner two thirds and outer one third interfaces of the ganglion cell layer and inner plexiform layer. Full RT was measured from the inner limiting membrane to the retinal pigment epithelium. Meanwhile, the entire macular zone (6 mm × 6 mm range) was divided into three concentric rings centered on the fovea (0–1 mm diameter fovea, 1–3 mm diameter parafovea, and 3–6 mm diameter perifovea) in the Early Treatment Diabetic Retinopathy Study (ETDRS). All images were captured under normal room illumination of 200 to 300 lx by the same experienced examiner between 8:00–11:00 AM. Images were rejected if the signal quality was less than 8. SRVD, DRVD, and RT were measured at baseline in the six- and seven-month follow-ups.

Axial Length and Cycloplegic Refraction Assessment

Axial length (AL) was measured before cycloplegia using a noncontact partial coherence interferometer (IOLMaster-500; Carl Zeiss Meditec AG, Jena, Germany). All SER measurements were obtained after a complete cycloplegic regimen (four drops of compound tropicamide eye drops: 0.5% tropicamide and 0.5% neosynephrine at an interval of 10-minute induced cycloplegia in both eyes). Refractive examinations were performed three times using an autorefractor (Topcon RM 8000 A; Topcon Medical Laser Systems, Livermore, CA, USA) 10 minutes after the last eye drop.

Statistical Analyses

Statistical analyses were performed using the Empower (R) software (www.empowerstats.com; X & Y Solutions Inc., Boston, MA, USA) and R (http://www.r-project.org). Only right-eye data from participants who completed the seven-month follow-up period were used in the analysis. Normally distributed continuous variables were expressed as mean ± standard deviation (SD) and were evaluated using Student's *t*-test. Skew-distributed continuous variables are presented as medians with first and third quartiles (Q1 and Q3) and were analyzed using the nonparametric rank-sum test. Categorical variables were expressed as percentages and evaluated using the chi-square test. Factors related to RVD and RT changes and differences in RVD and RT changes between the two groups were assessed using univariate and multivariate regression analyses. Multivariate regression analysis was used to adjust the confounding factors including sex, age, baseline AL and baseline RVD or RT of each corresponding zone. A P value <0.05 was considered statistically significant.

Results

Baseline Characteristics

A total of 69 eligible children were enrolled in this study (35 in the RLRL group and 34 in the 0.01% atropine group). After seven months, 62 children completed the study (32 in the RLRL group and 30 in the 0.01% atropine group (Fig.). No significant differences were found in the baseline parameters (age, sex, AL, SER, SRVD, DRVD, and RT) between the two groups at baseline or between the dropout participants and those who completed the study (all $P > 0.05$, Table 1).

Changes in SER and AL

After six months, the SER changes were −0.04 ± 0.17 D and −0.15 ± 0.24 D in the RLRL and 0.01% atropine groups, respectively ($P = 0.04$; Table 2). The corresponding AL changes were 0.08 ± 0.08 mm and 0.16 ± 0.14 mm in the two groups ($P = 0.01$; Table 2). In comparison, SER and AL changes in the placebo group (n = 25) of our previous study were −0.34 ± 0.35 D and 0.28 ± 0.14 mm, respectively. By comparing with the historical placebo group, the mean difference in the SER changes was 0.30 D and 0.19 D in the RLRL and 0.01% atropine groups, respectively,

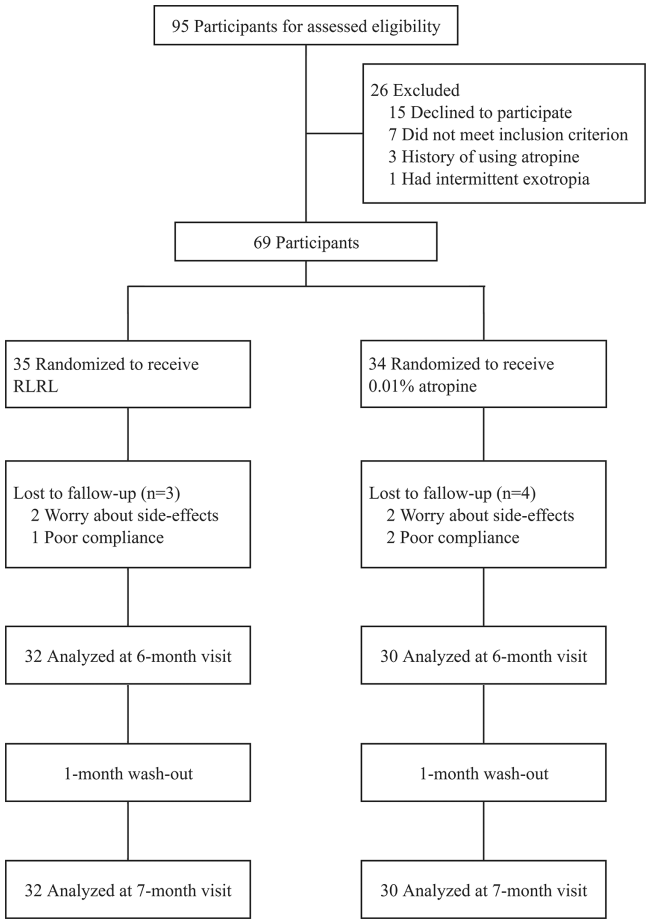


Figure. Participant recruitment and randomization flowchart.

and the mean difference in the AL changes were 0.20 mm and 0.12 mm in the above corresponding two groups. Both RLRL and 0.01% atropine effectively prevented myopic shift and axial elongation compared with placebo (all $P < 0.05$), and RLRL showed greater prevention of myopic shift and axial elongation than 0.01% atropine (all $P < 0.05$).

During the one-month recovery period from six months to seven months, the SER changes were −0.05 ± 0.05 D versus −0.06 ± 0.07 D and AL changes were 0.02 ± 0.06 mm versus 0.04 ± 0.05 mm in the RLRL and 0.01% atropine groups, there was no significant difference in the changes in SER and AL in either group and between the two groups (all $P > 0.05$).

Changes in SRVD, DRVD, and RT

Changes in SRVD, DRVD, and RT at six months are presented in Table 2. After six months, the whole, parafoveal, and perifoveal SRVD significantly increased in the RLRL and 0.01% atropine groups (all $P < 0.05$), and none of these changes varied significantly between the two groups (4.69% [1.92, 6.93] vs. 4.35% [0.96, 6.41] in whole, 5.97% [0.83, 8.61] vs. 3.92% [0.79, 6.62] in parafovea, 4.48% [2.13, 6.78] vs. 4.52% [1.04, 6.69] in perifovea; all $P > 0.05$), whereas foveal SRVD all remained stable in the two groups (all $P > 0.05$). In the RLRL group, the whole and perifoveal DRVD increased significantly (all $P < 0.05$), whereas no statistical difference was observed in the foveal and parafoveal DRVD changes (all $P > 0.05$). In the 0.01% atropine group, DRVD (whole, fovea, parafovea, and perifovea) remained stable (all $P > 0.05$). No significant differences were found in the changes in RT (including the whole, fovea, parafovea, and perifovea) between the two groups (all $P > 0.05$). SRVD and DRVD returned to baseline after stopping RLRL or 0.01% atropine for one month. Furthermore, no structural damage was observed in the choroid or retina using SS-OCTA data for all participants. After six months, the SRVD, DRVD and RT (foveal, parafoveal, perifoveal and whole) remained stable in the placebo group of our previous study (all $P > 0.05$).

Multivariate regression analyses after adjusting for confounding factors showed that the whole (adjusted sex, age, baseline AL, and baseline whole DRVD) and perifoveal DRVD (adjusted sex, age, baseline AL, and baseline perifoveal DRVD) in the RLRL group increased significantly more than that in the 0.01% atropine group ($\beta_{whole} = 2.57$, 95% CI: 0.70 to 4.45, $P < 0.01$; $\beta_{\text{perifovea}} = 2.86$, 95% CI = 0.87 to 4.84, $P < 0.01$); however, there were no significant differences in SRVD increase and RT changes (whole, fovea, parafovea and perifovea), and in foveal and parafoveal DRVD changes between the two groups (all $P > 0.05$; Table 3).

Factors Related to Changes in Whole SRVD and DRVD

In the RLRL group, univariate regression analysis showed that whole SRVD and DRVD changes were significantly negatively associated with whole baseline SRVD ($P = 0.01$, $P = 0.04$; Table 4) but not with baseline AL and SER, and AL and SER changes at six months. In the 0.01% atropine group, no factors were related to changes in the whole SRVD or DRVD (all $P > 0.05$; Table 4). Multivariate regression analyses adjusted for sex and age showed that the smaller the whole SRVD at baseline, the more the whole SRVD and DRVD increased after six months in the RLRL group ($\beta_{SRVD} = -0.42$, 95% CI = −0.81 to −0.04, $P = 0.04$; $\beta_{DRVD} = -0.39$, 95% CI = −0.77 to −0.01, $P = 0.04$).

Table 1. Baseline Characteristics of RLRL and 0.01% Atropine Groups

Variables	RLRL Group (n = 32)	0.01% Atropine Group (n = 30)	*P* Value
Age (yrs)	8.76 ± 1.24	8.79 ± 1.14	0.92
Female	17 (53.1%)	14 (46.7%)	0.61
SER (D)	−0.24 ± 0.29	−0.22 ± 0.30	0.79
AL (mm)	23.62 ± 0.77	23.59 ± 0.74	0.88
Superficial retinal vascular density			
Fovea	8.37% ± 3.20%	8.24% ± 3.55%	0.88
Parafovea	42.10% ± 5.16%	42.65% ± 2.83%	0.61
Perifovea	42.81% ± 3.58%	42.76% ± 2.49%	0.95
Whole	41.74% ± 3.38%	41.76% ± 2.39%	0.98
Deep retinal vascular density			
Fovea	20.01% ± 6.67%	19.24% ± 5.52%	0.62
Parafovea	52.19% ± 3.12%	52.88% ± 2.18%	0.32
Perifovea	51.99% ± 2.43%	52.22% ± 2.76%	0.73
Whole	51.64% ± 2.29%	51.44% ± 2.41%	0.74
Retinal thickness (µm)			
Fovea	244.57 ± 13.79	244.38 ± 16.26	0.96
Parafovea	330.09 ± 9.08	327.66 ± 10.41	0.33
Perifovea	290.08 ± 11.95	291.27 ± 8.36	0.65
Whole	297.81 ± 10.52	298.08 ± 8.24	0.91

Fovea: 0–1 mm diameter; parafovea: 1–3 mm diameter; perifovea: 3–6 mm diameter; whole: 0–6 mm diameter.

Table 2. Changes in Retinal Vascular Density, Retinal Thickness, SER, and AL Over Six Months, Mean ± SD or Median (Q1, Q3)

Variables	RLRL Group (n = 32)	0.01% Atropine Group (n = 30)	*P* Value
Superficial retinal vascular density (%)			
Fovea	−0.64 [−1.85, 0.74]	−0.14 [−0.92, 2.07]	0.21
Parafovea	5.97 [0.83, 8.61]*	3.92 [0.79, 6.62]*	0.35
Perifovea	4.48 [2.13, 6.78]*	4.52 [1.04, 6.69]*	0.76
Whole	4.69 [1.92, 6.93]*	4.35 [0.96, 6.41]*	0.60
Deep retinal vascular density (%)			
Fovea	0.32 [−1.13, 2.64]	1.13 [−1.09, 3.95]	0.74
Parafovea	2.82 [−1.48, 4.42]	−0.79 [−1.81, 2.68]	0.14
Perifovea	3.66 [0.55, 6.19]*	−0.39 [−1.41, 0.91]	0.01
Whole	3.22 [−0.05, 4.99]*	−0.37 [−1.33, 1.55]	0.01
Retinal thickness (µm)			
Fovea	0.29 [−2.06, 3.30]	0.17 [−1.77, 1.84]	0.40
Parafovea	1.48 [−0.47, 2.95]	0.04 [−0.74, 1.96]	0.24
Perifovea	1.44 [−1.15, 2.80]	−1.04 [−2.04, 1.74]	0.08
Whole	1.51 [−0.63, 3.07]	0.39 [−1.34, 1.82]	0.17
SER (D)	−0.04 ± 0.17	−0.15 ± 0.24	0.04
AL (mm)	0.08 ± 0.08	0.16 ± 0.14	0.01

Fovea: 0–1 mm diameter, parafovea: 1–3 mm diameter; perifovea: 3–6 mm diameter; whole: 0–6 mm diameter.
*Represents a significant difference between six months and baseline within the group ($P < 0.05$).

Table 3. Multivariate Regression Analyses of Changes in SRVD, DRVD, and RT in Two Groups

	Nonadjusted Model		Fully Adjusted Model	
Exposure Group	β (95% CI)	*P* Value	β (95% CI)	*P* Value
Changes in SRVD (%)				
Fovea				
0.01% atropine	0		0	
RLRL	−0.68 (−1.98 to 0.61)	0.30	−0.71 (−2.01 to 0.59)	0.28
Parafovea				
0.01% atropine	0		0	
RLRL	0.99 (−1.32 to 3.30)	0.40	−0.34 (−2.73 to 2.05)	0.78
Perifovea				
0.01% atropine	0		0	
RLRL	0.23 (−1.78 to 2.24)	0.82	−0.37 (−2.53 to 1.79)	0.73
Whole				
0.01% atropine	0		0	
RLRL	0.38 (−1.57 to 2.33)	0.70	−0.35 (−2.45 to 1.75)	0.74
Changes in DRVD (%)				
Fovea				
0.01% atropine	0		0	
RLRL	−0.20 (−2.14 to 1.73)	0.83	0.10 (−2.02 to 2.22)	0.92
Parafovea				
0.01% atropine	0		0	
RLRL	1.46 (−0.32 to 3.24)	0.11	1.88 (−0.06 to 3.81)	0.06
Perifovea				
0.01% atropine	0		0	
RLRL	2.96 (1.05 to 4.88)	<0.01	2.86 (0.87 to 4.84)	<0.01
Whole				
0.01% atropine	0		0	
RLRL	2.54 (0.78 to 4.31)	<0.01	2.57 (0.70 to 4.45)	<0.01
Changes in RT (μm)				
Fovea				
0.01% atropine	0		0	
RLRL	−0.06 (−2.72 to 2.59)	0.96	−0.12 (−3.20 to 2.95)	0.93
Parafovea				
0.01% atropine	0		0	
RLRL	1.51 (−0.72 to 3.73)	0.18	1.43 (−0.87 to 3.73)	0.22
Perifovea				
0.01% atropine	0		0	
RLRL	1.56 (−0.09 to 3.21)	0.07	1.25 (−0.41 to 2.92)	0.14
Whole				
0.01% atropine	0		0	
RLRL	0.89 (−0.55 to 2.33)	0.23	0.71 (−0.78 to 2.20)	0.35

Fovea: 0–1 mm diameter, parafovea: 1–3 mm diameter; perifovea: 3–6 mm diameter; whole: 0–6 mm diameter; Multivariate model is adjusted sex, age, baseline AL, and baseline RVD or RT of each corresponding zone.

Table 4. Univariate Regression Analyses of Changes in Whole SRVD and DRVD in Two Groups

	Changes in Whole SRVD				Changes in Whole DRVD			
	RLRL Group		0.01% Atropine Group		RLRL Group		0.01% Atropine Group	
Exposure	β (95% CI)	*P* Value	β (95% CI)	*P* Value	β (95% CI)	*P* Value	β (95% CI)	*P* Value
Sex	−2.03 (−4.54 to 0.48)	0.12	0.96 (−1.98 to 3.90)	0.53	−1.33 (−3.84, to 1.18)	0.30	0.31 (−1.95 to 2.57)	0.78
Age (yr)	0.38 (−0.31 to 1.06)	0.29	0.88 (−0.31 to 2.07)	0.16	0.16 (−0.51 to 0.82)	0.64	0.29 (−0.65 to 1.23)	0.54
Baseline AL	0.00 (−1.90 to 1.90)	0.99	2.01 (−0.43 to 4.45)	0.12	−0.52 (−2.38 to 1.33)	0.58	0.37 (−1.59 to 2.34)	0.71
Changes in AL	0.15 (−7.20 to 7.51)	0.96	2.45 (−8.45 to 13.34)	0.66	−6.16 (−13.07 to 0.74)	0.09	0.81 (−7.51 to 9.12)	0.85
Baseline SER	−2.10 (−5.12 to 0.93)	0.18	1.13 (−1.91 to 4.17)	0.47	0.93 (−2.10 to 3.96)	0.55	1.06 (−1.24 to 3.36)	0.37
Changes in SER	1.45 (−3.78 to 6.68)	0.59	−1.01 (−5.00 to 2.98)	0.62	2.24 (−2.88 to 7.36)	0.39	−1.85 (−4.80 to 1.10)	0.23
Baseline whole SRVD	−0.46 (−0.82 to −0.10)	0.01	0.09 (−0.54 to 0.72)	0.78	−0.36 (−0.69 to −0.02)	0.04	0.05 (−0.43 to 0.53)	0.84
Baseline whole DRVD	−0.41 (−0.97 to 0.15)	0.15	−0.33 (−0.94 to 0.28)	0.29	−0.23 (−0.79 to 0.33)	0.42	−0.29 (−0.75 to 0.17)	0.22
Baseline whole RT	0.05 (−0.08 to 0.17)	0.44	−0.00 (−0.18 to 0.18)	0.98	0.04 (−0.08 to 0.16)	0.51	0.03 (−0.11 to 0.17)	0.64

Discussion

This six-month prospective randomized trial compared the changes in SRVD, DRVD, and RT (including fovea, parafovea, perifovea, and whole) in premyopic children who received RLRL and 0.01% atropine. The results showed that SRVD increased in both groups at a similar magnitude and the DRVD was increased in the RLRL group but was stable in the 0.01% atropine group. There were no significant changes in RT in either group.

The present study showed a significant increase in SRVD (whole, parafovea, perifovea) and DRVD (whole and perifovea) in the RLRL group. Several studies have shown that RLRL positively affects children with myopia, including the slower growth of AL and SER.[3,18] However, the mechanism by which RLRL prevents myopia progression remains unknown. The retina plays a key role in myopia progression. A number of studies have shown that retinal blood flow and vascular density decrease in high myopia.[19,20] Retinal hypoxia causes an increase in the production of reactive oxygen species, which changes the levels of retinal dopamine and nitric oxide and further promotes myopia progression.[9] In the current study, retinal blood flow was significantly increased. Subsequently, it improved retinal hypoxia in the RLRL group, which may provide a new idea for studying the mechanism of myopia control. However, Yang et al.[11] reported that RVD increased after five minutes and decreased from five minutes to one hour when using RLRL in children with myopia. Zhao et al.[12] demonstrated that RVD remained unchanged after using RLRL for four weeks. There may be several reasons for the different results of RVD changes after using the RLRL. First, the power of the red light in the RLRL was different (0.16 mW vs. 1.07 – 1.42 mW vs. 0.85 mW in Yang's, Zhao's, and current studies, respectively). Zhou et al.[8] found that increases in subfoveal choroidal thickness differed after using a six-month RLRL at three different powers. We speculate that the trends in RVD may vary when different powers of red light are used in the RLRL in each study. Second, the follow-up time of the above two studies (five minutes to one hour and one month)[11,12] was relatively short compared with the current study (six months). Finally, RVD was scanned using different SS-OCTA modes. Zhao's study used a smaller scan range of 3 × 3 mm, including the fovea and parafovea but not the perifovea. Although Yang et al.[11] had a large scanning range of 18 mm × 18 mm, they used the average value of the nine regions of the ETDRS for analysis, which may have led to the loss of some information. The scanning range of the current

study was 6 mm × 6 mm, the RVD was divided into SRVD and DRVD, and the whole SRVD and DRVD were further divided into three concentric rings (fovea, parafovea, and perifovea). Both the whole and three concentric rings of SRVD and DRVD were analyzed in the current study.

SRVD (parafovea and perifovea) increased, whereas DRVD did not change in the 0.01% atropine group. Increasing evidence has shown that low-concentration atropine has excellent efficacy in delaying the progression of myopia in children.[21,22] However, the mechanism by 0.01% atropine controls myopia progression remains unclear. The choroid thickens after using atropine drops in children with myopia.[23,24] To date, studies on the effects of low-dose atropine on the retinal vasculature are scarce. This study showed an increase in SRVD but no change in DRVD in the 0.01% atropine group. However, Wang et al.[13] reported that children aged six to 14 years with SER < −6.0 D used one drop of 0.01% atropine every night for three months with no increase in both SRVD and DRVD. Although Wang et al.'s[13] and the current study adopted the same methods to stratify and partition RVD, the SS-OCTA instruments and their built-in algorithms for recognizing retinal blood flow differed.

In the present study, the RVD measurement and analysis methods were identical for the RLRL and 0.01% atropine groups. SRVD (whole, parafovea, and perifovea) increased similarly in both groups, and DRVD (whole and perifovea) increased in the RLRL group but not in the 0.01% atropine group. Additionally, no significant changes in SRVD and DRVD (whole, fovea, parafovea, and perifovea) after six months were found in the placebo group in our previous study. Data on changes in AL and SER in the placebo group have been published,[14] whereas the date from the SS-OCTA are not publicly available. Previously, the placebo group was recruited from the same population using the same inclusion criteria and had demographic and clinical parameters similar to those of the children in the current study. Therefore the changes in RVD in the RLRL and 0.01% atropine groups may be related to using RLRL or 0.01% atropine. The mechanisms of increased retinal blood flow after RLRL or 0.01% atropine are currently unknown and may be related to the release of nitric oxide (NO) in the retina. A study showed that the increase in skin microcirculation produced by polychromatic visible light is due to activation of NO synthesis.[25] Moreover, there is evidence from animal models suggesting that atropine promotes the release of dopamine and NO in the retina.[26] However, the SRVD changes (increased in both groups) were different from DRVD changes (increased in the RLRL group, but not in the 0.01% atropine group) after using RLRL and 0.01% atropine, which may be related to the different anatomical characteristics of SRVD and DRVD. The superficial retinal vessel is supplied by the central retinal artery and consists of capillaries of 75 mm in thickness, whereas the deep retinal vessel is supplied by vertical anastomoses from the superficial retinal vessel and consists of more complex and denser capillaries of smaller caliber.[27] Additionally, the reasons for the different variation trends in the SRVD and DRVD in the three concentric rings within and between the two groups remain unclear. However, SRVD and DRVD in both groups returned to baseline one month after discontinuation. Further studies are required to determine the reasons for the different effects on SRVD and DRVD after long-term RLRL or 0.01% atropine use and withdrawal. Moreover, the whole SRVD and DRVD changes after six months were negatively associated with the whole baseline SRVD, but not with changes in baseline AL, SER, and other factors in the RLRL group.

As in the placebo group, there was no significant change in RT both in the RLRL and 0.01% atropine groups after six months. Zhao et al.[12] and Wang et al.[13] also found no changes in RT after using RLRL for one month and 0.01% atropine for three months, respectively. The retina comprised a neurosensory layer and a pigment epithelium layer, along with blood vessels that supply them with nutrients. Zhao et al.[12] indicated that the evaluation of changes in RT helps evaluate photodamage. Photodamage can cause photoreceptor cells to die, reducing the thickness of the outer nuclear layer.[28,29] In the current study, no abnormal structural changes were observed in the SS-OCTA images in the RLRL and 0.01% atropine groups, indicating that both methods were safe for the retina. However, we should note the potential risks of fundus damage associated with RLRL laser exposure.[30] The current study also considered the potential risks of RLRL, and we strictly monitored the fundus structure at each follow-up visit, whereas no fundus structural change was found in one child during the follow-up in current study. Therefore more studies with longer follow-up are required to confirm the safety of RLRL therapy.

The current study had some limitations. First, we did not investigate the molecular mechanisms underlying these biological effects in animal models. Second, the observation period in this study was only six months, which may not be long enough to observe the full impact of RLRL and 0.01% atropine on RVD and thickness. We are conducting a long-term follow-up study to provide future results.

In conclusion, the present study found that SRVD increased with a similar magnitude in the RLRL and

0.01% atropine groups, whereas DRVD increased only in the former group; there were no significant changes in RT and no documented functional or structural damage in either group after six months of treatment in premyopic children. Further research is required to understand the long-term effects and potential molecular mechanisms.

Acknowledgments

The authors thank Xiaohui Li, Engineer of Chenzhou Eye Care Health Technology Co. Ltd for his support of the RLRL instrumentation technology.

Supported by Key R&D and Promotion Project of Henan Science and Technology Department (grant no. 201801591); Key Scientific Research Project of Universities of Henan Education Department (grant no. 19A320066). New technology and new project of the First Affiliated Hospital of Zhengzhou University (grant no. 2022-C50).

Disclosure: **L. Shang**, None; **S. Gao**, None; **W. Wang**, None; **M. Chang**, None; **N. Ma**, None; **C. Huang**, None; **S. Yu**, None; **M. Wang**, None; **A. Fu**, None

[*] LS and SG contributed equally to this work and share first authorship.

References

1. Holden BA, Fricke TR, Wilson DA, et al. Global prevalence of myopia and high myopia and temporal trends from 2000 through 2050. *Ophthalmology*. 2016;123:1036–1042.
2. Yam JC, Jiang Y, Tang SM, et al. Low-Concentration Atropine for Myopia Progression (LAMP) study: a randomized, double-blinded, placebo-controlled trial of 0.05%, 0.025%, and 0.01% atropine eye drops in myopia control. *Ophthalmology.* 2019;126:113–124.
3. Yam JC, Li FF, Zhang X, et al. Two-year clinical trial of the Low-Concentration Atropine for Myopia Progression (LAMP) Study: Phase 2 Report. *Ophthalmology.* 2020;127:910–919.
4. Jiang Y, Zhu Z, Tan X, et al. Effect of repeated low-level red-light therapy for myopia control in children: a multicenter randomized controlled trial. *Ophthalmology*. 2022;129:509–519.
5. Xiong R, Zhu Z, Jiang Y, et al. Longitudinal changes and predictive value of choroidal thickness for myopia control after repeated low-level red-light therapy. *Ophthalmology*. 2023;130:286–296.
6. Chiang ST, Turnbull PRK, Phillips JR. Additive effect of atropine eye drops and short-term retinal defocus on choroidal thickness in children with myopia. *Sci Rep*. 2020;10(1):18310.
7. Hao Q, Zhao Q. Changes in subfoveal choroidal thickness in myopic children with 0.01% atropine, orthokeratology, or their combination. *Int Ophthalmol*. 2021;41:2963–2971.
8. Zhou W, Liao Y, Wang W, et al. Efficacy of different powers of low-level red light in children for myopia control. *Ophthalmology.* 2024;131:48–57.
9. Francisco BM, Salvador M, Amparo N. Oxidative stress in myopia. *Oxid Med Cell Longev*. 2015;2015:750637.
10. Tkatchenko TV, Tkatchenko AV. Pharmacogenomic approach to antimyopia drug development: pathways lead the way. *Trends Pharmacol Sci*. 2019;40:833–852.
11. Yang W, Lin F, Li M, et al. Immediate effect in the retina and choroid after 650 nm low-level red light therapy in children. *Ophthalmic Res.* 2023;66:312–318.
12. Zhao C, Ni Y, Zeng J. Effect of red-light therapy on retinal and choroidal blood perfusion in myopic children. *Ophthalmic Physiol Opt*. 2023;43:1427–1437.
13. Wang Y, Zhu X, Xuan Y, et al. Short-term effects of atropine 0.01% on the structure and vasculature of the choroid and retina in myopic Chinese children. *Ophthalmol Ther*. 2022;11:833–856.
14. Wang W, Zhang F, Yu S, et al. Prevention of myopia shift and myopia onset using 0.01% atropine in premyopic children—a prospective, randomized, double-masked, and crossover trial. *Eur J Pediatr*. 2023;182:2597–2606.
15. Yu S, Du L, Ji N, et al. Combination of orthokeratology lens with 0.01% atropine in slowing axial elongation in children with myopia: a randomized double-blinded clinical trial. *BMC Ophthalmol*. 2022;22:438.
16. Cheng D, Ruan K, Wu M, et al. Characteristics of the optic nerve head in myopic eyes using swept-source optical coherence tomography. *Invest Ophthalmol Vis Sci*. 2022;63(6):20.
17. Lang Y, Kwapong WR, Kong L, et al. Sparser macula microvasculature in neuromyelitis optica spectrum disorder occurs independently of optic neuritis. *Mult Scler Relat Disord*. 2022;58:103470.

18. Zhou L, Xing C, Qiang W, et al. Low-intensity, long-wavelength red light slows the progression of myopia in children: an Eastern China-based cohort. *Ophthalmic Physiol Opt*. 2022;42(2):335–344.
19. Al-Sheikh M, Phasukkijwatana N, Dolz-Marco R, et al. Quantitative OCT angiography of the retinal microvasculature and the choriocapillaris in myopic eyes. *Invest Ophthalmol Vis Sci*. 2017;58(4):2063–2069.
20. Lin F, Zhao Z, Li F, et al. Longitudinal macular retinal and choroidal microvasculature changes in high myopia. *Invest Ophthalmol Vis Sci*. 2021;62(15):1.
21. Chia A, Lu QS, Tan D. Five-year clinical trial on atropine for the treatment of myopia 2: myopia control with atropine 0.01% eyedrops. *Ophthalmology*. 2016;123:391–399.
22. Cui C, Li X, Lyu Y, et al. Safety and efficacy of 0.02% and 0.01% atropine on controlling myopia progression: a 2-year clinical trial. *Sci Rep*. 2021;11(1):22267.
23. Ye L, Shi Y, Yin Y, et al. Effects of atropine treatment on choroidal thickness in myopic children. *Invest Ophthalmol Vis Sci*. 2020;61(14):15.
24. Zhao W, Li Z, Hu Y, et al. Short-term effects of atropine combined with orthokeratology (ACO) on choroidal thickness. *Cont Lens Anterior Eye*. 2021;44(3):101348.
25. Samoilova KA, Zhevago NA, Petrishchev NN, et al. Role of nitric oxide in the visible light-induced rapid increase of human skin microcirculation at the local and systemic levels: II. healthy volunteers. *Photomed Laser Surg.* 2008;26: 443–449.
26. Feldkaemper M, Schaeffel F. An updated view on the role of dopamine in myopia. *Exp Eye Res.* 2013;114: 106–119.
27. Samara WA, Say EA, Khoo CT, et al. Correlation of foveal avascular zone size with foveal morphology in normal eyes using optical coherence tomography angiography. *Retina.* 2015;35:2188–2195.
28. Boulton M, Rozanowska M, Rozanowski B. Retinal photodamage. *J Photochem Photobiol B*. 2001;64(2–3):144–161.
29. O'Steen WK, Donnelly JE. Chronologic analysis of variations in retinal damage in two strains of rats after short-term illumination. *Invest Ophthalmol Vis Sci.* 1982;22:252–255.
30. Liu H, Yang Y, Guo J, et al. Retinal damage after repeated low-level red-Light laser exposure. *JAMA Ophthalmol.* 2023;141:693–695.